

DESIGN AND ANALYSIS OF ALGORITHMS

Practical No. 9: Bellman-Ford Algorithm

1. Aim

To find shortest paths from a single source in a weighted graph using Bellman-Ford and detect negative-weight cycles.

2. Objective

- Understand repeated edge relaxation.
- Find shortest paths when negative edge weights are present.
- Detect a reachable negative-weight cycle.

3. Theory

Bellman-Ford is a single-source shortest path algorithm that supports negative edge weights. It repeatedly relaxes every edge $V - 1$ times for a graph with V vertices. One additional pass checks whether any distance can still be reduced; if so, a reachable negative-weight cycle exists. Its time complexity is $O(VE)$.

4. Conclusion

Bellman-Ford computes shortest paths in graphs containing negative edge weights and can detect reachable negative-weight cycles. The practical demonstrates the importance of repeated relaxation.